Figure 1 (draft): joint inflammation in RA and the pathways that repurposed drugs target

A. In the RA joint

DCs

recruit and activate

T cells

stimulate

differentiate

B cells

Macrophages

Synoviocytes (FLS)

aggressive, tumour-like

Chondrocytes

Osteoclasts

IL-6 · IL-21 · IL-23
→ STAT3 → Th17 differentiation

NF-κB → IL-1β, IL-6, TNF-α

TNF-α · IL-6 · IL-1β · MMPs (also IL-17, IL-21, IL-23)

TNF-α, IL-6

TNF-α and IL-6R antagonists (clinical use)

Synovial cell proliferation
Cartilage and bone damage

RANKL (TNF superfamily)

Osteoclast differentiation

RA synovial cells: Ang II ↑ RANKL via AT1R;
AT1R blocker (olmesartan) prevents it, AT2R blocker does not [cells]

B. Intracellular pathways and repurposed-drug entry points

JAK/STAT

FDA-approved; baricitinib (JAK1/2) approved for RA

JAK inhibitors

STAT3 activation ↓, IL-21 ↓; may reduce Th17 differentiation [cells]

Fenofibrate

NF-κB

linked to JNK/NF-κB signalling in FLS [cells]

Dulaglutide

proliferation, metabolism, angiogenesis, survival

PI3K/AKT

MAPK · WNT · SYK

Metformin

AMPK

activates AMPK in hepatocytes [cells]

p-AMPK/t-AMPK ↑ in AIA rat joints [animal]

Dapagliflozin

raptor phosphorylation, energy stress [cells]

six metabolic pathways (glycolysis, TCA, PPP, FAO, FAS, amino acids) act in RA progression;
CD4+ T: glycolysis → PPP;
FLS: anaerobic glycolysis, lactate ↑

mTORC1

mitophagy ↑ → inflammasome ↓, IL-1β, IL-18 ↓ [abstract];
TNF-α-induced FLS: IL-1β, IL-6, IL-8 ↓, IL-10 ↑ [cells]

PPAR-γ reduced in RA FLS [cells]

PPAR-γ

Pioglitazone

HMG-CoA reductase

IFN-γ-induced MHC-II ↓; L-mevalonate reverses [cells]

Statins

AT1R

experimental arthritis ameliorated [animal]

Losartan

activates / increases · inhibits / decreases · link between panels

[cells] in vitro · [animal] rodent model · [abstract] abstract only · no patient data in panel B

Grey boxes = approved therapies. DCs, dendritic cells; FLS, fibroblast-like synoviocytes; MMPs, matrix metalloproteinases; PPP, pentose phosphate pathway; FAO/FAS, fatty acid oxidation/synthesis; AIA, adjuvant-induced arthritis.